Advanced Machine Learning Project

1 Data Description and Pre-Processing

The Autotrader dataset consists of 402,005 used-vehicle listings and 12 predictor variables. The aim is to develop a regression model that predicts selling prices based on vehicle characteristics. The features include numerical variables (price, mileage, year_of_registration), categorical variables (standard_make, standard_model, standard_colour, vehicle_condition, body_type, fuel_type), a binary indicator (crossover_car_and_van). Identifier fields (public_reference, reg_code) were removed because they do not carry predictive value and can introduce noise into the modelling process.

The price variable shows extreme right skew, ranging from £120 to £9.9 million (Figure 1.1, left panel). To stabilise variance without discarding observations, winsorisation was applied at the 1st and 99th percentiles and stratified by vehicle condition (new/used). This procedure reduced the maximum to £96,201 while lowering the standard deviation by 34% (from £21,900 to £14,463). Outcome statistics (Figure 1.1, right panel): mean £16,435, median £12,600.

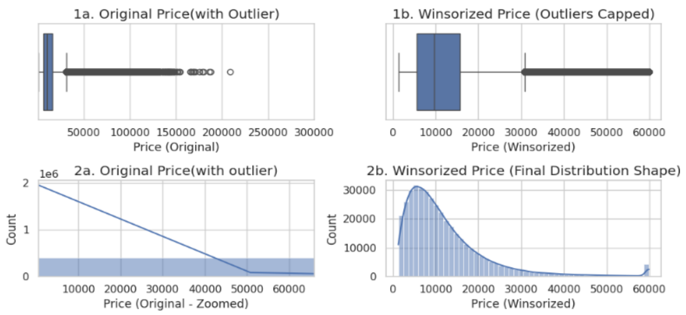

Figure 1.1. Price distribution before and after winsorisation.

Mileage also showed significant positive skew, with most vehicles below 50,000 miles and a long upper tail. A log-transformation was applied to stabilise variance and reduce the influence of high-mileage outliers. During the modelling stage, mileage was log-transformed to reduce variance distortion. Vehicle age was derived from year_of_registration. This feature showed a strong inverse relationship with price, reflecting natural depreciation patterns. Spearman correlations (Figure 1.2) indicated that vehicle age ($\rho = -0.71$) and log_mileage ($\rho = -0.65$) were the strongest numerical predictors of price among the continuous features.

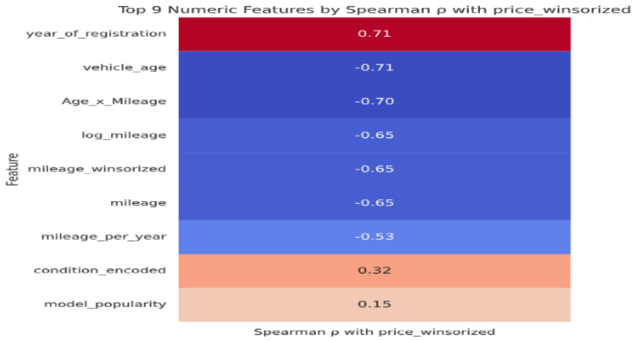

Figure 1.2 Spearman Correlation Heatmap of Numerical Features.

Categorical features display meaningful price separation. For example, SUVs and electric or hybrid cars tend to fall into higher price ranges, while hatchbacks and petrol or diesel cars cluster in lower segments. High-cardinality features, particularly make and model, were target-encoded to avoid the high dimensionality created by one-hot encoding.

Missing values were imputed using domain-knowledge-based imputation. The year_of_registration variable had 33,311 missing entries. These were imputed using domain-aware rules based on mileage, vehicle condition, and UK registration patterns. Remaining gaps were filled using mapped DVLA registration codes. Missing mileage values were imputed using grouped medians by make, model, and year, with a global median applied as a fallback. Missing standard_colour values were imputed using the modal colour within each manufacturer.

Feature engineering was applied to improve the predictive signal. The Vehicle_age feature was obtained from the year of registration, giving a clearer picture of how the value drops over time. Mileage got log transformed, as it was heavily skewed to the right when checked in data exploration. The mileage_per_year feature (mileage / Vehicle_age) was engineered to capture usage intensity, distinguishing lightly-used older vehicles from heavily-used ones.

Categorical encoding was applied according to feature characteristics. High-cardinality variables were target-encoded to keep price information while avoiding dimensionality issues. Low-cardinality nominal variables were one-hot encoded, and ordinal features were encoded according to their natural ranking. All preprocessing was implemented within a ColumnTransformer pipeline to ensure consistent transformations across training and test sets and to prevent data leakage.

2. **Automated Feature Selection (AFS)**

The dataset includes numerical features, several one-hot encoded categorical features, and two high-cardinality variables (standard_model and standard_make). With around 400k rows and mixed feature types, some predictors are weak or redundant. Automated Feature Selection was used to remove features that add noise, reduce overfitting, improve training speed, and make the final model easier to understand.

All selection and modelling were done on the pre-processed matrices. Preprocessing included imputation, scaling for numeric columns, target encoding for the high-cardinality columns, and one-hot encoding for categorical variables. Train, validation, and test sets were created using stratified splits on price quintiles, so the target distribution remained consistent.

The first method of AFS selected was SelectFromModel with ExtraTrees. This model is a meta transformer that helps machine learning models, such as a decision tree or logistic regression, to help make decisions. It looks at how much each feature helps the model make a prediction and keeps only the ones that meet a certain importance level.

A random subsample of 30,000 training was fitted with ExtraTrees for fast and sufficiently stable learning, it learned importance and transformed train, validation, and test using the importance features. This method was applied because it captures nonlinear relationships and interactions between features that are important for vehicle price prediction. The selection kept the strongest feature importance, for example, mileage, the age of the car, the make and model encoding and key body types and reduced dimensionality with less loss of information. This means it shows strong performance with mixed numeric and categorical encoding.

The second method applied is Recursive feature Elimination with cross-validation (RFECV) using a ridge estimator. RFECV fit models repeatedly, find and remove the feature with the lowest weight or weakest features, and stop when validation performance stops improving. In our dataset and its size, PCA keeping the runtime is important. RFECV was fit on an 8k subsample of the training matrix with step = 0.1, cv=3 and a lower bound of 30 features. RFECV selected 35 out of 38 processed features, showing that most features still contribute useful information. The resulting mask was applied to the train, validation, and test sets. The reason for applying this method is that there is no guessing; instead, guessing how many features to keep, RFECV uses cross-validation to choose the number of features. Also, it is a reliable score as it uses ridge, it stays calm even if the data is messy or has similar columns, and it ignores noise.

Three regressors were trained on both feature sets: Random Forest, HistGradientBoosting, and Ridge.

Random Forest gave the best results overall:

The best performer overall was RF.

- SFM (ExtraTrees): Val MAE £636.99, Test MAE £640.87, R^2 = 0.9877
- RFECV (Ridge): Val MAE £795.93, Test MAE £801.69, R^2 = 0.9843

RFECV removed various features, for instance, the categorical feature, especially fuel type, and the popularity class one-hot. These are vital for tree splits, so the accuracy dropped by £160 MAE.

HistGradientBoosting:

- SFM: Test MAE £1087.63, R^2 0.9801
- RFECV: Test MAE £1085.92, R^2 0.9802

AFS present similar performance, which denotes robustness to minor changes in feature selections.

Ridge (linear baseline):

- SFM: Test MAE £3791.00, R^2 0.8304
- RFECV: Test MAE £3760.69, R^2 0.8328

The result demonstrates that the Ridge performed poorly compared to the other two models, overall, with an MAE of £3,700 because of the strong nonlinearity in car prices. RFECV worked slightly better than SFEM for the model, which is expected for a linear model.

Overall observation, across all the models, validation and testing metrics were closely aligned, meaning there is no overfitting.

Final choice:

SelectFromModel (SFM) with ExtraTrees outperformed Recursive Feature Elimination (RFECV) with Ridge as the primary automated feature selection method. When used with the RF model, it gave the most accurate result; it only missed the price by about £640, while RFECV missed £796. The reason SFM performed better than RFECV was that SFM knows that vehicle price is complicated; for instance, car brand matters more if the mileage is low. It kept those features together. However, RFECV looks for straight lines. It threw one of the key features, like fuel type, as it did not see a simple, straight-line relationship to price. This made the RF model less accurate by 160.

3. Tree Ensembles

Tree ensembles are methods that combine multiple decision trees into a single dependable model. This method is used widely, as relying on one tree can be unstable and easily overfit. The idea is to create a group of trees and let them work together so that the combination of many models can perform better.

Decision trees alone are not enough because it is simple and interpretable, but they suffer from high variance, meaning minor changes in the training data can lead to quite different trees; they can overfit easily if they grow too deep, and underfit if they are heavily regularised.

Tree ensembles are used because they reduce the problem by aggregating predictions from many trees, which reduces the variance and improves generalisation. These models are highly effective for structured datasets, for example, our dataset, car pricing, because it captures nonlinear relationships and feature interaction without needing a strong assumption about the data distribution.

For the baseline modelling, three approaches were evaluated: Random Forest, Gradient Boosting and Ridge Regression. These baselines were trained using the features selected by the SFM (ExtraTrees) method. Each model was fitted on the processed training data and evaluated using MAE on both the validation and test sets. The Random Forrest builds 100 decision trees; each trained on a random subset of the data and features. The final predication is the average of all trees. The Gradient boosting builds trees sequentially, and each new tree learns to correct the errors of the previous trees. It creates 100 boosting stages. The Ridge regression is a linear regression with regularisation; it is not a tree model, however, it is included as a baseline comparison, as it is required to find if advanced models outperform simple models.

Table 1 demonstrates that Ridge performed significantly worse than both tree-based models, achieving a test MAE of £3790.99 compared with £640.87 for Random Forest and £1933.81 for Gradient Boosting. This clearly shows that the nonlinear structure captured by the ensemble models is important for modelling car prices and that linear methods like ridge could not match the predictive capability of the tree-based baseline enough.

	Model	Test_MAE	Test_R2
0	Random Forest (baseline)	640.87	0.99
1	Gradient Boosting (baseline)	1933.81	0.94
2	Ridge (baseline)	3791.00	0.83
3	RF (tuned)	272.23	0.99
4	GB (tuned)	1553.67	0.95

Table 1. Baseline model performance on the validation and test sets.

Hyperparameter tuning was applied using GridSearchCV to increase and enhance the model's performance. RandomisedSearchCV with stratified subsampling of 5,000 training examples was used to reduce computational cost while preserving parameter sensitivity. For the random forest, the search analysed and explored differences in maximum tree depth, minimum leaf size, and feature sampling ratios. Gradient Boosting was tuned using the HistGradientBoostingRegressor (HGBR) because it is much faster and more efficient, as it uses histogram-based binning. Also, our dataset is large; it scales better to a large dataset, and it also manages missing values natively, simplifying the turning workflow. The mean absolute error was used as the cross-validation scoring metric to make sure the tuning was aligned with the evaluation goal.

All the models were evaluated on the validation set, and the best performance was retrained and refitted on the full training data set before final testing. Random Forest presented the greatest improvement, as it reduced its test MAE from £640.87 to £272.23, as shown in Table 1. Gradient Boosting also demonstrate improvement but remains less accurate, with a test MAE of £1553.67. A clear comparison is presented in Table 1, where the tuned Random Forest clearly appears as the strongest model. The results show that ensemble tree methods, specifically Random Forest, acquire the complex price relationships in the dataset far more effectively than a linear model like Ridge.

To figure out what the tuned RF model learned, feature importances were extracted and shown in Figure 3.1. he target-encoded feature standard_model was by far the most significant feature, taking half of the total importance. In the automotive industry, this is expected as the model of the car identity strongly determines base market value, and target encoding captures historical pricing patterns. Some engineered mileage-based features, for example, price_per_mile, log_mileage, and mileage_per_year, also ranked highly. These features capture vehicle usage and depreciation more effectively than raw mileage, which explains their strong predictive contribution. However, categorical features such as fuel type contributed quite little, suggesting that once model identity and mileage patterns are included, fuel type adds minimal additional explanatory value.

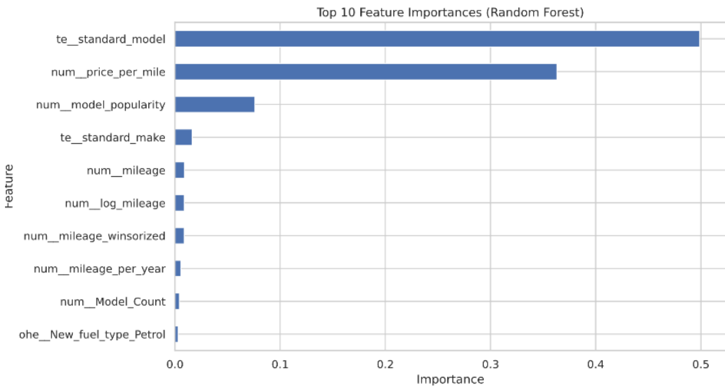

Figure 3.1. A bar chart showing the top 10 features' importance (Random Forest).

4. Ensemble of Tree Ensembles

Ensemble methods merge the predictions of many models and create a stronger model that has improved accuracy, stability, and robustness. After tuning the RF and HGBR model, three ensembles method ware used and explored, these are simple averaging, weighted averaging, and stacking. The objective was to investigate whether combining two strong tree-based learners could make further performance gain beyond the best current model (RF).

Simple Average

Predictions from RF and HGBR were merged with equal weight. This approach decreases variance but treats both base models equally accurately. On the validation and test sets, simple averaging attained MAEs of £874.11 and £868.88, with an R-squared of 0.9810, as shown in Table 2. It clearly shows this ensemble performed better than HGBR alone and underperformed compared with RF. HGBR prediction reduced the accuracy of the stronger model, which leads to no improvement over the RF baseline.

Weighted Averaging

What weighted averaging does is assign different importance to each model's prediction. Weights were improved to reduce validation MAE. Table 2 presents the resulting weights RF = 1.0 and HGBR = 0.0; this means the ensemble successfully selected Random Forest as the dominant model. This produced MAEs of £272.56 validation and £272.23 test with an R-squared of 0.9921. Compared to the rest of the performance values across all ensemble approaches, this is the best performance shown in Table 2. The results denote that Random Forest captures nonlinear relationships and feature dynamics far more effectively than HGBR, adding small benefits in the dataset.

Stacking Regressor

A StackingRegressor was applied on the dataset, using HGBR and RF as the base learners and Ridge regression as the meta learner. This method was used to increase predictive ability by combining all three models through the process, which mitigates individual model weaknesses and reduces errors. The meta learner was trained on cross validation n predication from the base models. This method achieved MAEs of £773.03 on validation and £769.78 on test, with an R-squared of 0.9823, as shown in Table 2. The result suggested that although stacking combines model predication, in this dataset, the meta learner could not outperform the chosen single model because of random forest dominance, ridge linear combination has limiting added benefit.

	Model	Val MAE	Test MAE	Test R²
0	Avg (RF + HGBR)	874.11	868.88	0.9810
1	Weighted Avg (w_rf=1.00, w_gb=0.00)	272.56	272.23	0.9921
2	StackingRegressor (RF+GB → Ridge)	773.03	769.78	0.9823

Table 2. Comparison of simple averaging, weighted averaging, and stacking ensembles using tuned RF and HGBR models

The results of Table 2 demonstrate that weighted averaging is the only ensemble method that selected the most accurate model, which is Random Forest, as it shows it achieved the Mean Absolute Error and the highest R-squared. The simple averaging reduced predictive performance by including the weaker HGBR, while the stacking regressor added small benefits because of the strong dominance of Random Forest. This clearly suggests that weighted ensembles are the most successful and effective when the base model performance varies significantly, and that careful selection of the model is important for ensemble success.

5. Feature Importance

Feature importance analysis was conducted; the Random Forest model was selected as the best-performing model based on its successful evaluation results. RF provides a built-in feature importance score; it calculates how much each feature reduces prediction error across all decision trees. Two feature importance approaches were applied; the models built in feature importance and permutation importance calculated on a small subset validation. Using both feature importances offers a more reliable indication of feature relevance.

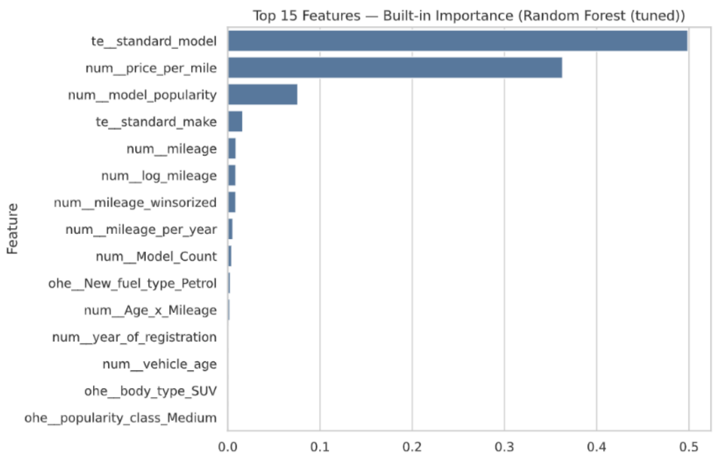

Figure 4.1. A bar plot showing the top 15 features ranked by the Random Forest's built-in importance scores.

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Top 10 (Built-in importance):
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	feature	importance
0	te__standard_model	0.498890
1	num__price_per_mile	0.363065
2	num__model_popularity	0.076070
3	te__standard_make	0.016443
4	num__mileage	0.009095
5	num__log_mileage	0.008856
6	num__mileage_winsorized	0.008822
7	num__mileage_per_year	0.005612
8	num__Model_Count	0.004316
9	ohe__New_fuel_type_Petrol	0.002944

Table 3 shows the top 10 ranked by the Random Forest's built-in importance scores.

Built-in Feature Importance

Figure 4.1 and Table 3 show the top features ranked by the RF built-in importance score. The target encoded feature (te_standard_model) has the highest importance, accounting for half of the total importance (0.498). This denotes that the model of the vehicle is the most influential feature for predicting vehicle prices in machine learning models, which implies that, with domain expectation, different models are popular and have different market demand, which dictates base price and resale value.

Table 3 also shows the next most important feature, which is the numeric price per mile, highlighting the strong impact of intense usage on price, for example, how much value is left in the car relative to how hard it has been driven. Other influential features include num_model_popularity and te_standard_make, implying that market demand and brand reputation also have a huge, meaningful impact on the pricing. For example, popular brand like Ford Fiesta and Volkswagen Golf sells faster and often hold more predictable prices than luxury and high-performance cars. Table 3 also presents that mileage-based features such as num_mileage, num_log_mileage, and num_mileage_winsorized appear lower in the ranking but still provide useful information by capturing depreciation patterns. In comparison, categorical features like fuel type are the least influential as it contributes little once model identity and usage features are included.

Permutation Importance

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Running SAFE permutation importance on 1500 rows...
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	feature	importance_mean	importance_std
5	num__price_per_mile	17433.835933	382.067644
12	te__standard_model	2858.048298	240.336657
0	num__mileage	1897.778187	31.253487
2	num__mileage_winsorized	1864.948724	34.404738
4	num__log_mileage	1858.442931	27.347986
10	num__model_popularity	1004.952228	81.935855
7	num__mileage_per_year	744.212850	14.127794
11	te__standard_make	660.548717	58.339699
9	num__Age_x_Mileage	434.859952	8.976064
8	num__Model_Count	128.869153	11.534409

Table 4. Shows Permutation Importance Scores for Vehicle Price Prediction Features.

Table 4 summarises permutation importance analysis and confirms that the models rely most heavily on the calculated ratio and feature engineering rather than raw data. Shuffling the feature num_price_per_mile produced by far the largest increase in error when permuted, making it the most important predictor under this evaluation. This suggests that the relationship between mileage usage and price is crucial to the model's predictive performance.

The target encoded standard model is still highly influential as it runs increased in MAE when shuffled. Similarly, for mileage-related features like num_mileage, num_mileage_winsorized, num_log_mileage, and num_mileage_per_year, they also rank highly, denoting that the model influences numerous representations of mileage to capture nonlinear depreciation effects. te_standard_make and num_model_popularity features also contributed; however, interaction terms and low-variance categorical features show insignificant impact.

6. SHAP/PDP Model Explanations

To improve the transparency and interpretability of the chosen model, RF, Shapley Additive explanations (SHAP), and Partial Dependence Plot (PDP) were used to analyse how features influence the pricing predictions. It has chosen RF because it achieved the strongest predictive performance; it is necessary to use these interpretability methods to understand its decision-making process and confirm that predictions are driven by meaningful relationships.

SHAP Analysis

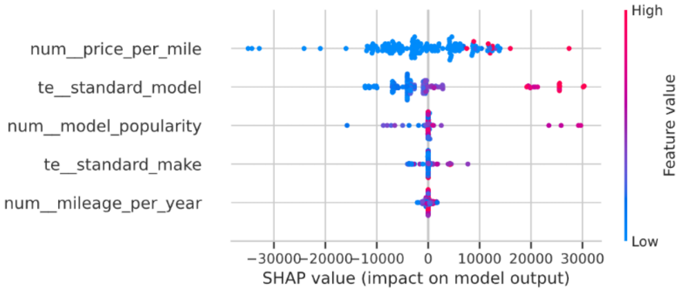

Figure 6.1. A summary plot showing SHAP values, impact on model output.

Figure 6.1 displays the SHAP summary plot for the RF model. The plot shows the top 5 distribution of SHAP values for the most significant features, where each represents an individual prediction and the colour denote that features value from low to high significance.

AS shown in Figure 6.1, the feature num_price_per_mile displays the widest spread of SHAP value, showing as the most influential predictor. The red shown in the higher values of prices per mile increases the prediction strongly upward, while the blue lower value reduces the predicted price. Again, this clearly denotes that vehicle usage efficiency plays a crucial and central role in determining the market values.

The second feature, the te_standard_model, also displays a strong influence with SHAP value spreading both positive and negative ranges. This shows the price difference between the models of the cars, implying the importance of model identity in price. The rest of the features, such as num_model_popularity, te_standard_model, and num_mileage_per_year, influence the predictions but with smaller importance. Overall, the SHAP analysis confirms that the model relies on a small set of dominant related features.

Partial Dependence Plots

To further analyse, understand and visualise the interaction between the prices and features, PDPs were used and generated.

Figure 6.2 illustrates a clear, gradual increase in the predicted price across different standard model categories, suggesting that changes in car model correspond to different shifts in baseline price. The plot shows a sharp jump at 25,000, which is around higher encoded values, and it confirms that luxury and high-performance car models are associated with significantly higher predicted prices.

Figure 6.2. A plot showing the PD for the standard model.

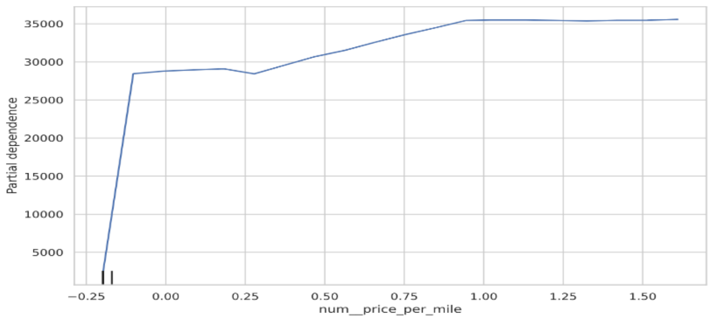

Figure 6.3. A plot showing PDP for num_price_per_mile.

Figure 6.3 exhibits that the relationship is strongly nonlinear; for example, predicted price increases as the price per mile increases at lower values before it gradually flattens at high levels above 30,000. The PDP clearly demonstrates that an increase in price per mile influences the price at low values; however, this effect weakens at higher values. This nonlinear pattern justifies why tree-based models, such as RF, outperform linear regression for these methods.

As proven in the figures, SHAP and PDP provide good analysis and insights into how the model makes predictions. For example, SHAP explains 5 separate predictions by showing how much features influence the last price, and the PDP displays how the predicted price changes as single features increase or decrease.

These methods consistently show standard model and mileage-related features as the main drivers of price. The result denotes that the RF model learns a clear and realistic pricing trend rather than relying on random and misleading relationships, which ensures confidence in both the accuracy and reliability of the final model.

7. Dimensionality Reduction (Linear)

Dimensionality reduction aims to reduce the number of features while keeping most of the information. High-dimensional data often has highly correlated or unnecessary features, which can increase model complexity and reduce interpretability. In this section, Principal Component Analysis (PCA) were used and generated as a linear dimensionality reduction technique to transform the original feature space into a smaller set of components that captures most of the variance in the car data.

PCA is sensitive to the scale of the dataset, so it was standardised using "StandardScaler" to ensure all the features had a zero mean and unit variance. Before applying PCA, this step is important. PCA was then fitted to the standardised training data to determine the principal components and the proportion of variance explained by each component.

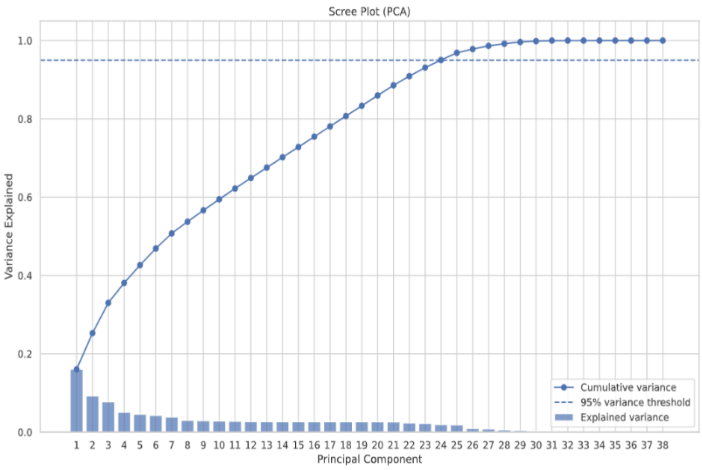

Figure 7.1. A plot showing the scree plot and cumulative explained variance from the PCA.

The plot shows the first few Principal components display noticeable variance, with the first component contribution the largest share. The increase in cumulative variance is gradual rather than steep, suggesting that variance is spread across many components rather than concentrated in just a few features.

As showed in Figure 7.1, 95% of the cumulative variance is reached after around 25 principal components, decreasing the dimensionality from the original 38 features to a smaller set of transformed variables. Figure 7.2 illustrates this gradual accumulation of variance, proving that many components are needed to retain most of the information contained in the original dataset.

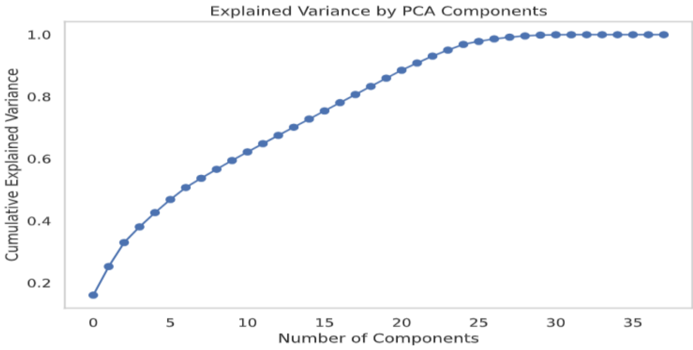

Figure 7.2. A plot showing the gradual accumulation of variance.

Despite the reduction features, the regression model trained (ridge) on PCA-transformed features gave weaker predictive performance. The PCA-based model achieved a validation MAE of approximately £4549 and a test R-squared of around 0.77, which is extensively lower than the performance achieved using the original features.

The result of the figures and the PCA denotes that although dimensionality was successfully decreased, the key predictive information was lost during the transformation. PCA preserves variances but does not necessarily keep features that are most useful for prediction. In the AutoTrader dataset, the original variables capture specific and meaningful pricing trends that are weakened when converted into a linear combination of principal components.

Also, tree-based models, for example, RF and Gradient Boosting, can normally manage high-dimensional data and complex feature relationships. Based on the results achieved, applying linear dimensionality reduction provided less benefit and can negatively impact predictive performance. It is seen that the figures suggest that retaining the original feature-engineered features is more effective than using PCA for this predictive section.

8. Dimensionality Reduction (Non-Linear)

Non-linear dimensionality reduction methods were used to investigate whether complex trend patterns exist in the vehicle dataset that are not captured by linear techniques like PCA. PCA shows and provides a linear combination of features, but methods like t-SNE and Isomap can show structure driven by feature relationships. These techniques were applied and compared with PCA to assess whether car prices show meaningful structure in a low-dimensional space.

To apply these methods, all features were standardised to make sure consistent scaling. Because computation is expensive and costly so random sample of 5,000 observations from the training set was used. The three methods applied include PCA as a linear baseline, t-SNE with two components and a perplexity of 30 to keep local neighbourhood structure, and Isomap with two components and 15 neighbours to keep global manifold structure. The results were visualised with a scatter plot with points coloured by vehicle price.

The reasons for this approach are not predictive; it is for exploratory analysis to understand whether price can be represented in a lower-dimensional space and to explain the robust performance of nonlinear models. Using scatter plots and visual analysis helps to influence whether price variation can be explained by a low-dimensional representation and to understand very well why a nonlinear model, for example, RF, achieves robust performance on the dataset.

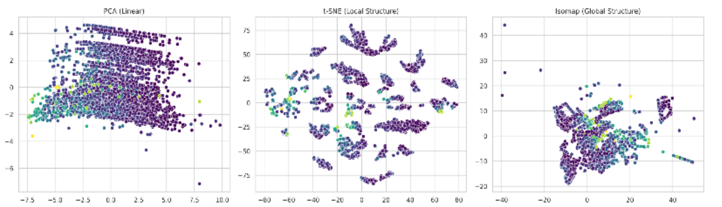

Figure 8.1. A scatter plot showing a comparison of the estimates created by PCA, t-SNE and Isomap.

As shown in Figure 8.1, on the left-hand side of the graph, the PCA shows a widely dispersed cloud with strong overlap across the price values, suggesting no clear separation is visible by price. However, in the middle graph, the t-SNE shows strong, clear local clustering, with nearby points of similar colours showing close together. This indicates that cars that have a comparable price range tend to form local neighbourhoods, suggesting the presence of a nonlinear structure in the dataset. The plot shows no clear global ordering and structure, but the clusters are well separated locally.

On the right-hand side, the plot displays an Isomap projection, and it shows a smoother global structure compared to PCA, but the price values still overlap and remain mixed across regions. Figure 8.1 clearly denotes that Isomap keeps broader relationships better than PCA; the lack of strong price separation shows that global structure alone does not fully explain the price difference.

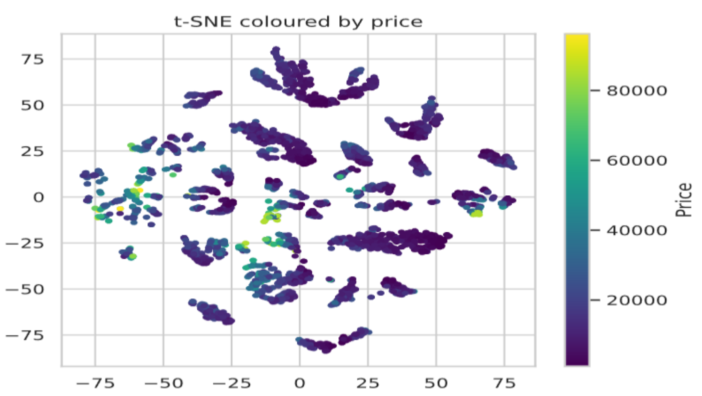

Figure 8.2. A plot showing the t-SNE projection coloured by car price.

As shown in Figure 8.2, t-SNE further supports these analyses and insights. Even though there is no clear linear gradient present, local regions of the plots tend to display consistent colour patterns, suggesting that nearby points often share a comparable price range or level.

Overall, these results denote that the AutoTrader dataset has meaningful nonlinear local structure, which seems not well captured by linear dimensionality reduction methods, for example, PCA. Both in Figure 8.1 and 8.2, the strong local clustering observed in t-SNE clarifies why nonlinear, tree-based models, for example, RF, perform well, because they naturally capture local relationships between features.

However, the fact that clear separation by price implies that the car price is influenced by multiple interacting variables rather than one single low-dimensional factor. This suggests why nonlinear dimensionality reduction methods like t-SNE and Isomap are useful for visual exploration but not suitable for feature compression or predictive modelling in this section.

9. Polynomial Regression

Polynomial regression is a form of linear regression; it extends models by adding interaction terms between features. This allows the model to capture nonlinear relationships while remaining a parametric method. For this section, polynomial regression with Ridge regularisation was used to assess whether model feature relationships and interaction could improve predictive performance compared to standard linear regression and tree-based models like RF.

To implement the polynomial regression, a pipeline consisting of two stages was used. First was Polynomial feature expansion, followed by Ridge regression for regularisation. To control computational complexity, only interaction terms were included (interaction_only=True) and polynomial degrees of 1 and 2 were evaluated. Degree 1 relates to standard ridge regression, and degree 2 relates to and introduces pairwise feature interactions.

To improve the efficiency, a subsample of 20,000 observations was taken from the training data and used to reduce computational cost. Models were trained and evaluated on the full validation and test sets using MAE and R-squared. Higher degrees were not tested because of the rapid growth in feature dimensionality and computational cost.

The goal of this analysis and this section was to evaluate whether manually adding nonlinear interaction can improve the linear model and decrease the performance gap with tree-based techniques, for example, RF. Polynomial regression gives a controlled way to create nonlinearity while keeping a simple model structure.

```
Training Polynomial Degree 1...
Training Polynomial Degree 2...
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	Model	Val_MAE	Test_MAE	Test_R2
0	Poly degree 1	3808.18	3811.27	0.83
1	Poly degree 2	2739.25	2723.80	0.91

Table 5. shows a comparison of two different polynomial degrees.

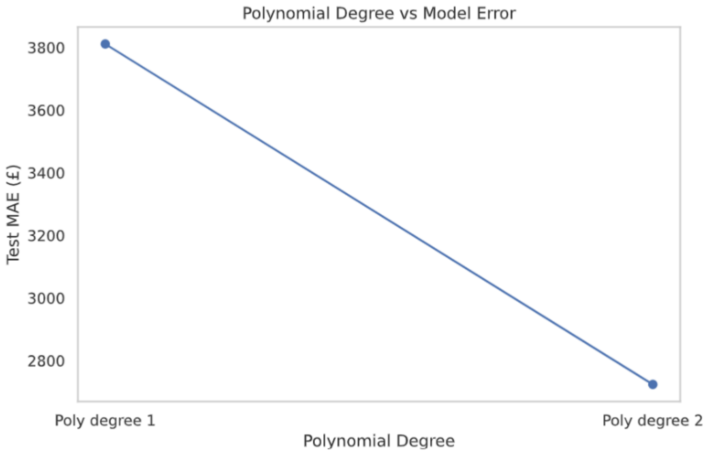

Figure 9.1. A plot showing the difference between the polynomials of degree 1 and 2.

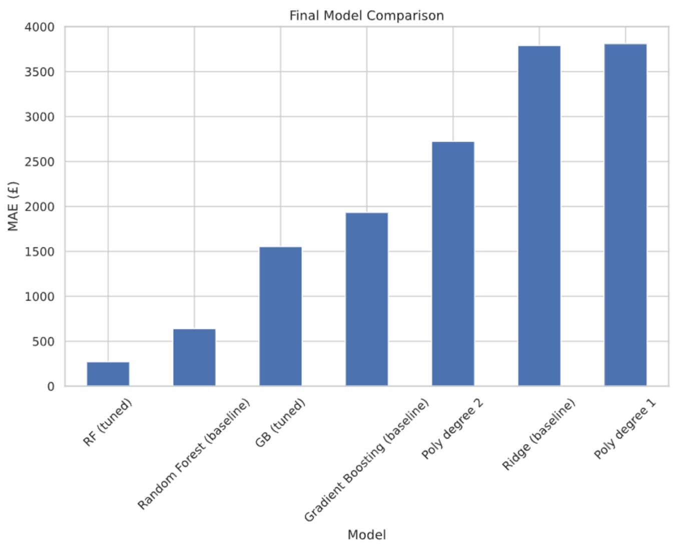

Figure 9.2. A bar plot showing the final models' comparison.

Table 5 demonstrates the performance of polynomial regression models with degrees 1and 2. Poly degree 1, which relates to the standard ridge regression, achieved a test MAE of £3811.27 and an R-squared of 0.83, suggesting weak predictive performance. This proves that linear models struggle to capture the complete complexity of the dataset.

Poly degree 2 shows a noticeable improvement, because the model reduced test MAE to £2723.80 and increased R-squared to 0.91, suggesting that the relationship between features contributes useful predictive information. Figure 9.1 clearly shows this reduction in error when moving top to bottom (from degree 1 to degree 2).

When compared with other models in Figure 9.2, polynomial regression stays less accurate than ensemble tree methods. Figure 9.2 also shows that the model performs far worse than the tuned random forest, as it achieved a test MAE of £272.23, and underperforms gradient boosting models.

The figures and results denote that polynomial regression can improve performance over simple linear regression by capturing some feature relationships. But those will be small and limited. Polynomial models rely on constructed relations, and they rapidly become computationally expensive because the dimensionality of the features increases, which makes their ability to model complex relationships harder to efficiently.

The big performance gap between polynomial regression and tree-based ensembles suggests the weaknesses of polynomial regression for this task. RF and Gradient Boosting quickly learn complex, nonlinear, and higher-order relations directly from the dataset without needing manual feature increase. This makes them far more effective and scalable for modelling vehicle prices.

Overall, the results and figures show that Polynomial regression offers some benefits over linear models, but it is not competitive with ensemble tree methods. Figure 9.2 clearly shows that the tuned Random Forest stays the strongest model, confirming that flexible, non-parametric approaches are best suited to capturing the complex structure of the dataset.

10. Clustering for Feature Engineering

The last section, clustering, was applied as a feature engineering technique. Instead of predicting with clusters, the cluster label was added as an extra categorical feature to the model. The aim was to see if grouping similar vehicles could provide matching information to a supervised learning model and improve predictive performance.

K-means clustering was applied to the training data after the feature selection. To reduce computation time and costs, a sample of. A sample of 10,000 rows was used to fit the clustering model, then the number of clusters was set to k = 8.

When fitted, cluster labels were predicted for the full training, validation, and test sets. The labels were then encoded with one-hot and added to the original features. The RF model was trained on the new dataset using a sample of 50,000 rows for efficiency. The model Performance was evaluated on validation and test data.

The reason and idea for clustering-based feature engineering are that clusters could capture complex relationships between vehicles that are not easily shown by individual features. For example, if vehicles in the same cluster have similar prices, then the cluster feature might help the model make better predictions.

Figure 10.1. A histogram showing cluster distribution on the training set.

Based on the observations, the histogram in Figure 10.1 shows that some clusters are much larger than others, which means an unevenly populated dataset. This suggests that the dataset has natural grouping and certain types of vehicles appear more often, for example, VW Golf, Ford Focus, Vauxhall Corsa) small and hatchback cars.

	Model	Test_MAE	Test_R2
0	Random Forest (baseline)	640.87	0.99
1	Gradient Boosting (baseline)	1933.81	0.94
2	Ridge (baseline)	3791.00	0.83
3	RF (tuned)	272.23	0.99
4	GB (tuned)	1553.67	0.95
5	Poly degree 1	3811.27	0.83
6	Poly degree 2	2723.80	0.91
7	RF + Cluster Feature	524.37	0.98

Table 5. shows the final comparison table and model results.

Table 5 compares the performance of the RF model with the cluster features against the previously evaluated model. RF with cluster feature achieved a test MAE of £524.37 and a test R-squared of 0.98, which denotes good improvement over the baseline RF but stays weaker than the tuned Random Forest model, which attained a test MAE of £272.23. When compared across all models, the clustering improved model ranks better than linear, polynomial, and baseline ensemble methods, but does not outperform the best-tuned tree-based methods.

The results denote that clustering did not help a lot but introduced some useful information. This is because RF already learns similar grouping on its own by splitting the data into regions, which is like clustering, so the cluster features add little information. However, the clusters are not clearly linked to price and are uneven; this means they do not stand for strong price groups.